21-242 Recitation 12

Linear Transformations — Q6. Injectivity iff the image of a basis is independent

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that $T : V \to W$ is injective iff $T(\vec{b}_1), \ldots, T(\vec{b}_n)$ is linearly independent.

Course-level solution

($\Rightarrow$). Suppose T is injective and $\sum a_i T(\vec{b}_i) = \vec{0}$. Then $T\left(\sum a_i \vec{b}_i\right) = \vec{0}$, so $\sum a_i \vec{b}_i \in \ker T = \{\vec{0}\}$ (injective linear maps have trivial kernel: if $T(\vec{v}) = T(\vec{w})$ then $T(\vec{v} - \vec{w}) = \vec{0}$, so $\vec{v} - \vec{w} = \vec{0}$). Independence of $\mathcal{B}$ forces every $a_i = 0$. Thus $\left(T(\vec{b}_i)\right)$ is independent.
($\Leftarrow$). Suppose $\left(T(\vec{b}_i)\right)$ is independent and $T(\vec{v}) = \vec{0}$. Write $\vec{v} = \sum a_i \vec{b}_i$. Then $\sum a_i T(\vec{b}_i) = \vec{0}$, so every $a_i = 0$, hence $\vec{v} = \vec{0}$. Thus $\ker T = \{\vec{0}\}$, so T is injective.

Later-course solution (using ideas from the rest of the handout)

After rank-nullity: T is injective iff $\dim \ker T = 0$ iff $\dim \operatorname{im}(T) = \dim V = n$ iff the n vectors $T(\vec{b}_i)$ are independent in W (an independent list of length $\dim \operatorname{im}(T)$ is a basis of the image).